BAYESIAN INVERSION FOR PDE MODELS IN tnreason

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

We represent all solutions of a PDE to different parameters as tensor networks. This enables the integration of logical constraints and formulation of inference tasks, such as sampling from Bayesian posteriors, based on contractions.

1 PDE solutions as tensors

Given a discretization of a function, we investigate the indicator tensor, whether a combination of values is a solution to the PDE.

1.1 Tensor representation of solutions

First of all we discretize continuous functions towards a tensor representation:

- Selection of d domain elements
- Selection of m image elements together with an approximator mapping each image element to one of the selected
- Variables X_k representing the function evaluation at the $k \in [d]$ domain element

In this discretization scheme any functions is represented by an index $x_{[d]}$. A set of function is represented by its basis encoding

$$\tau^{\mathcal{F}}\left[X_{[d]} = x_{[d]}\right] = \begin{cases} 1 & \text{if} \quad x_{[d]} \in \mathcal{F} \\ 0 & \text{else}\,. \end{cases}$$

We are interested in the discretized solution sets $\mathcal{F}$ of PDEs and in decompositions of $\tau^{\mathcal{F}}\left[X_{[d]}\right]$. Here, we include parameters of the PDE as further variables Z.

Example 1 (One-dimensional Darcy flows)**.** *In the example of one-dimensional Darcy flows, the variables X_k describe the pressure at a position k and further variables Y_k the medium resistance at that position.*

1.2 Euler method

With Eulers method a solution to a PDE is approximated step by step. Given the value x_k the x_{k+1} is computed given the parameters z. We define f_k as the function mapping x_k and z to the corresponding x_{k+1}.

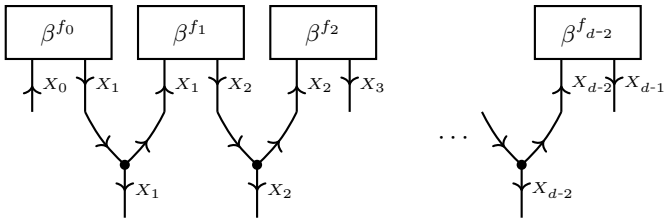
The set of Euler approximations $\mathcal{F}$ of the solutions to a PDE is then decomposed as

$$\tau^{\mathcal{F}}\left[X_{[d]}, Z\right] = \left\langle \{\beta^{f_k}\left[X_{k+1}, X_k, Z\right] \,:\, k \in [d-1]\} \right\rangle_{\left[X_{[d]}, Z\right]}\,.$$

Example 2 (Toy example)**.** *Consider the parameter free toy example $u' = 1$ on the interval $[0, 1]$ discretized into d equal size cells. We choose $m \geq d$ and enumerate $[m]$ another discretization of an interval with equal cell size. The discretized solution is then captured by the functions*

$$f_k(x_k) = x_k + 1$$

and the solutions tensor is the decomposition:



1.3 Generalization to Runge Kutte methods

The generalization to Runge-Kutte methods of higher orders p is straight forward: Each formula f_k computing x_{k+1} requires further arguments $x_k, \ldots, x_{k-p+1}$. As a consequence, the decomposition of $\tau^{\mathcal{F}}$ involves tensors of larger orders, namely

$$\tau^{\mathcal{F}}\left[X_{[d]}, Z\right] = \left\langle \left\{ \beta^{f_k}\left[X_{k+1}, X_k, \ldots, X_{k-p+1}, Z\right] \,:\, k \in [d-1] \right\} \right\rangle_{\left[X_{[d]}, Z\right]} .$$

2 Integration of logical constraints

Example 3 (Constrained parameters in Darcy flows)**.** *When assumptions are made on the medium variables in Darcy flows, they can be added as further tensors.*

3 Discretization of Gaussians

We investigate a discretization scheme for Gaussians, which preserves the maximum entropy property with respect to the sufficient statistic.

3.1 Standard Gaussians

A Gaussian distribution on $\mathbb{R}$ with mean y and standard derivation σ is the distribution

$$\begin{aligned}\mathbb{P}\left[X = x\right] &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-y)^2}{2\sigma^2}\right] \\ &= \exp\left[-\frac{1}{\sigma^2}x^2 + \frac{y}{\sigma^2}x - \frac{y^2}{\sigma^2} - \ln\left[\sqrt{2\pi\sigma^2}\right]\right] .\end{aligned}$$

We notice that it is in the exponential family to the statistic $t(x) = (x, x^2)$ with the canonical parameter

$$\theta\left[L\right] = \begin{pmatrix} \frac{y}{\sigma^2} \\ -\frac{1}{\sigma^2} \end{pmatrix}\left[L\right]$$

and the mean parameter (can be verified by integration)

$$\mu\left[L\right] = \begin{pmatrix} y \\ \sigma^2 + y^2 \end{pmatrix}\left[L\right] .$$

The mean polytope is the area bounded by a parabola (the image set of the statistic). The zero-noise limit are the maximum entropy distributions on the parabola, namely the Dirac distributions to the corresponding values.

3.2 Discretization

Let us now restrict the values $\mathbb{R}$ to a finite set $\mathcal{U} \subset \mathbb{R}$, which is enumerated by a variable X taking values in $[m]$. We define discrete Gaussian distributions on $\mathcal{U}$ as members of the exponential family with sufficient statistic $t(x) = (x, x^2)$. The mean polytope is now the convex hull of the two-dimensional vectors on the parabola which are images of $\mathcal{U}$.

3.3 Multivariate Gaussians

The discretization scheme preserving the maximum entropy property can be extended to multivariate Gaussians. The sufficient statistics is then $t(x_{[d]}) = (x_{[d]}, x_{[d]}x_{[d]}^T)$, where by $x_{[d]}x_{[d]}^T$ we denote the unfolded matrix of product of the indices.

4 Noisy observations

Let us consider access to observations of the solution function at specific domain elements. We capture this by adding tensor cores to each observation.

When the observations of each values are independent, e.g. discretized Gaussian noise, they are added by elementary networks. The corresponding sufficient statistics are x_k and $(x_k)^2$ to each observed value and the Gaussian noise is a maximum entropy distribution.

4.1 Zero-noise limit

The zero-noise limit can be investigated based on the characterization of generic maximum entropy distributions Goessmann and Eigel (2026).

Example 4 (Observations in the toy example)**.** *In the setting Example 2 let us discretize the solution with $d = 2$, i.e. only two values are considered. Let us consider observations of both by y_0 and y_1. If $y_1 = y_0 + 1$, then the zero-noise limit is $\epsilon_{y_0,y_1}[X_0, X_1]$. The more involved limit is $y_1 \neq y_0 + 1$, in which the observations are not agreeing with respect to any solution of the PDE. In this case, we can apply the characterization of maximum entropy distributions Goessmann and Eigel (2026) to characterize the limit.*

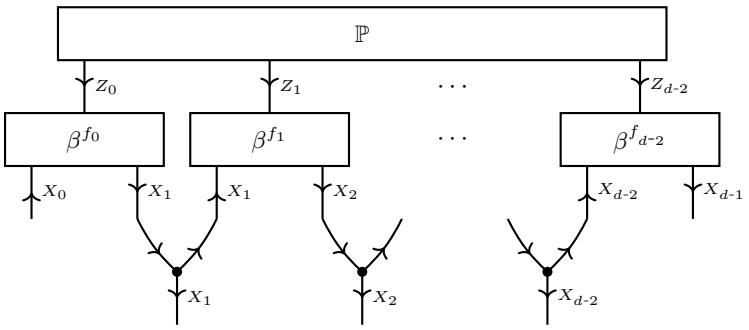
If $y_1 = y_0 + 2$ the limit is the uniform distribution on the support of the corresponding face, namely

$$\frac{1}{2} \cdot (\epsilon_{y_0,y_0+1}[X_0, X_1] + \epsilon_{y_0+1,y_0+2}[X_0, X_1]) \ .$$

5 Stochastic Differential Equations

We generalize to stochastic differential equations by considering probability distributions over parameters.

Example 5 (Eulers method with random parameters)**.** *Consider an ODE discretized to $X_{[d]}$ with parameters enumerated by $Z_{[d-1]}$. Eulers method gives the next function value x_{k+1} based on the current x_k and parameter Z_k. We denote f_k as function computing these Euler steps. When we have a distribution $\mathbb{P}$ of the parameters, then the Euler approximated solution of the SDE is the contraction:*



References

Alex Goessmann and Martin Eigel. Tensor network representations of discrete maximum entropy distributions via mean polytopes, September 2026. URL http://arxiv.org/abs/2609.07184. arXiv:2609.07184 [math.ST].